

《多元函数微分学》检测题参考解答

一、(15分) 1. B 2. D 3. A 4. C 5. C

二、(15分)

1. e

2. $2x - y - z - 1 = 0$.

3. $f_{12} - \frac{y}{x^2} f_{22}$.

4. $\sin x(1+x^2)^{\sin x-1} \cdot 2x + (1+x^2)^{\sin x} \ln(1+x^2) \cdot \cos x$.

5. $\frac{x+2}{25} = \frac{y-1}{28} = \frac{z-6}{12}$.

三、(10分) 设函数 $f(u)$ 具有连续导数, $z = f(e^x \cos y)$ 满足 $\cos y \frac{\partial z}{\partial x} - \sin y \frac{\partial z}{\partial y} = (4z + e^x \cos y)e^x$. 若

$f(0) = 0$, 求 $f(u)$ 的表达式.

解: 因为 $z = f(e^x \cos y)$,

所以 $\frac{\partial z}{\partial x} = f'(e^x \cos y)e^x \cos y$, $\frac{\partial z}{\partial y} = -f'(e^x \cos y)e^x \sin y$. (4分)

$\cos y \frac{\partial z}{\partial x} - \sin y \frac{\partial z}{\partial y} = f'(e^x \cos y)e^x \cos^2 y + f'(e^x \cos y)e^x \sin^2 y = f'(e^x \cos y)e^x$, (5分)

由已知 $f'(e^x \cos y)e^x = (4z + e^x \cos y)e^x$, 即 $f' - 4f = e^x \cos y$. (7分)

令 $u = e^x \cos y$, 即 $f'(u) - 4f(u) = u$. 由一阶线性非齐次微分方程式知:

$f(u) = e^{\int 4du} \left(\int u e^{-\int 4du} du + C \right) = -\frac{u}{4} - \frac{1}{16} + C e^{4u}$. (9分)

又因为 $f(0) = 0$ 可得 $C = \frac{1}{16}$. 所以 $f(u) = -\frac{u}{4} - \frac{1}{16} + \frac{1}{16} e^{4u}$. (10分)

四、(10分) 设 $z(x, y)$ 是由方程组 $\begin{cases} x = (t+1)\cos z, \\ y = t \sin z \end{cases}$ 确定的隐函数, 求 $\frac{\partial z}{\partial x}$.

解: 在方程组两边将 x, y 视为自变量, 将 z, t 视为隐函数, 方程组两边对 x 求偏导数, 有

$$\begin{cases} \cos z \frac{\partial t}{\partial x} - (t+1) \sin z \frac{\partial z}{\partial x} = 1, \\ \sin z \frac{\partial t}{\partial x} + t \cos z \frac{\partial z}{\partial x} = 0 \end{cases} \quad (8分) \quad \text{解得: } \frac{\partial z}{\partial x} = \frac{-\tan^2 z}{y + x \tan^3 z}. (10分)$$

五、(10分) 设 $f(x,y) = \begin{cases} (x^2+y^2)\sin\frac{1}{x^2+y^2}, & x^2+y^2 \neq 0 \\ 0, & x^2+y^2 = 0 \end{cases}$, 证明: (1) $f(x,y)$ 在 $(0,0)$ 点连续; (2)

$f_x(0,0), f_y(0,0)$ 存在; (3) $f_x(x,y), f_y(x,y)$ 在 $(0,0)$ 点不连续;

(4) $f(x,y)$ 在 $(0,0)$ 点可微.

解: (1) $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} (x^2+y^2)\sin\frac{1}{x^2+y^2} = 0 = f(0,0)$, $f(x,y)$ 在 $(0,0)$ 点连续. (2分)

(2) $f_x(0,0) = \lim_{x \rightarrow 0} \frac{f(x,0) - f(0,0)}{x} = \lim_{x \rightarrow 0} \frac{x^2 \sin \frac{1}{x^2} - 0}{x} = \lim_{x \rightarrow 0} x \sin \frac{1}{x^2} = 0$; 类似 $f_y(0,0) = 0$. (4分)

(3) $\therefore \frac{\partial f}{\partial x} = \begin{cases} 2x \sin \frac{1}{x^2+y^2} - \frac{2x}{x^2+y^2} \cos \frac{1}{x^2+y^2}, & x^2+y^2 \neq 0 \\ 0, & x^2+y^2 = 0 \end{cases}$ (5分)

$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\partial f}{\partial x} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} (2x \sin \frac{1}{x^2+y^2} - \frac{2x}{x^2+y^2} \cos \frac{1}{x^2+y^2})$ 不存在 (6分)

($\because$ 若取 $y=x$ 的路径趋于 0, $\lim_{\substack{x \rightarrow 0 \\ y=x}} 2x \sin \frac{1}{2x^2} = 0$, 但 $\lim_{\substack{x \rightarrow 0 \\ y=x}} \frac{1}{x} \cos \frac{1}{2x^2}$ 不存在)

$\therefore \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\partial f}{\partial x} \neq f_x(0,0) = 0$, 即 $\frac{\partial f}{\partial x}$ 在 $(0,0)$ 点不连续, 类似可得 $\frac{\partial f}{\partial y}$ 在 $(0,0)$ 点也不连续. (7分)

(4) $\therefore \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{\Delta f - A\Delta x - B\Delta y}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{[(\Delta x)^2 + (\Delta y)^2] \sin \frac{1}{(\Delta x)^2 + (\Delta y)^2}}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} \\ = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \sqrt{(\Delta x)^2 + (\Delta y)^2} \sin \frac{1}{(\Delta x)^2 + (\Delta y)^2} = 0, \text{ (9分)} \quad \therefore f(x,y) \text{ 在 } (0,0) \text{ 点可微. (10分)}$

六、(10分) 在曲面 $2x^2 + y^2 - z^2 - 2xy + 1 = 0$ 上求一点, 使它到原点的距离最小.

解: 令 $F(x,y,z,\lambda) = x^2 + y^2 + z^2 + \lambda(2x^2 + y^2 - z^2 - 2xy + 1)$ (4分)

令 $\begin{cases} F_x = 2x + 4\lambda x - 2\lambda y = 0 \\ F_y = 2y + 2\lambda y - 2\lambda x = 0 \\ F_z = 2z - 2\lambda z = 0 \\ F_\lambda = 2x^2 + y^2 - z^2 - 2xy + 1 = 0 \end{cases}, \text{ (8分)} \quad \text{解得 } x=0, y=0, z=\pm 1 \text{ (10分)}$

七、(10分) 设 $z = f(x+y, xe^y)$, 其中 f 具有二阶连续偏导数, 求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, \frac{\partial^2 z}{\partial x \partial y}$.

解: $z = f(x+y, xe^y)$, $\frac{\partial z}{\partial x} = f_1 + f_2 e^y$, (2分) $\frac{\partial z}{\partial y} = f_1 + f_2 x e^y$, (4分)

$$\frac{\partial^2 z}{\partial x \partial y} = f_{11} + f_{12} x e^y + e^y f_2 + e^y (f_{21} + x e^y f_{22}) \quad (8分) = f_{11} + (x+1)e^y f_{12} + x e^{2y} f_{22} + e^y f_2. \quad (10分)$$

八、(10分) 设 $F(u, v)$ 具有连续的一阶偏导数, 求证: 曲面 $F\left(\frac{x-a}{z-c}, \frac{y-b}{z-c}\right) = 0$ 上任一点处的切平面

都通过一定点 (其中 a, b, c 为常数)

证明: 设切点为 (x, y, z) , 切平面的法向量为 $\vec{n} = (F_x, F_y, F_z)$, (2分)

$$F_x = \frac{F_1}{z-c}, \quad F_y = \frac{F_2}{z-c}, \quad F_z = -\frac{x-a}{(z-c)^2} F_1 - \frac{y-b}{(z-c)^2} F_2 \quad (6分)$$

过点 (x, y, z) 的切平面方程为:

$$\frac{F_1}{z-c}(X-x) + \frac{F_2}{z-c}(Y-y) - \left[\frac{x-a}{(z-c)^2} F_1 + \frac{y-b}{(z-c)^2} F_2 \right] (Z-z) = 0 \quad (8分)$$

$\therefore X=a, Y=b, Z=c$ 满足方程, $\therefore$ 任一切平面都过定点 (a, b, c) . (10分)

九、(10分) 设变换 $\begin{cases} u = x + a\sqrt{y} \\ v = x + 2\sqrt{y} \end{cases}$ 把方程 $\frac{\partial^2 z}{\partial x^2} - y \frac{\partial^2 z}{\partial y^2} - \frac{1}{2} \frac{\partial z}{\partial y} = 0$ 化为 $\frac{\partial^2 z}{\partial u \partial v} = 0$, 试确定 a . 其中 $z(x, y)$

具有二阶连续偏导数.

解: $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v},$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{a}{2\sqrt{y}} + \frac{\partial z}{\partial v} \cdot \frac{1}{\sqrt{y}} = \frac{1}{\sqrt{y}} \left(\frac{a}{2} \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right), \quad (2分)$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial u^2} + 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}, \quad (4分)$$

$$\frac{\partial^2 z}{\partial y^2} = -\frac{1}{2} y^{-\frac{3}{2}} \left(\frac{a}{2} \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) + \frac{1}{\sqrt{y}} \left(\frac{\partial^2 z}{\partial u^2} \cdot \frac{a^2}{4\sqrt{y}} + \frac{\partial^2 z}{\partial u \partial v} \cdot \frac{a}{\sqrt{y}} + \frac{\partial^2 z}{\partial v^2} \cdot \frac{1}{\sqrt{y}} \right), \quad (6分)$$

$$\text{代入方程 } \frac{\partial^2 z}{\partial x^2} - y \frac{\partial^2 z}{\partial y^2} - \frac{1}{2} \frac{\partial z}{\partial y} = 0,$$

$$\Rightarrow \frac{\partial^2 z}{\partial x^2} - y \frac{\partial^2 z}{\partial y^2} - \frac{1}{2} \frac{\partial z}{\partial y} = \left(1 - \frac{a^2}{4} \right) \frac{\partial^2 z}{\partial u^2} + (2-a) \frac{\partial^2 z}{\partial u \partial v} = 0, \quad (9分)$$

$$\Rightarrow \begin{cases} 1 - \frac{a^2}{4} = 0 \\ 2-a \neq 0 \end{cases}, \Rightarrow a = -2. \quad (10分)$$